AMERICAN HEART ASSOCIATION (AHA) & AMERICAN COLLEGE OF CARDIOLOGY (ACC)

Blood Pressure Classification Table & Standardized Measurement Protocol

2017 ACC/AHA High Blood Pressure Clinical Practice Guidelines in Adults

Official Reference: 2017 ACC/AHA Guideline (Hypertension. 2018;71:e13-e115)

Official Web Portal: https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-r

1. The 5 Official AHA/ACC Blood Pressure Categories

Category	Systolic (SYS)	Operator	Diastolic (DIA)	Clinical Recommendation
Normal	Less than 120 mmHg	AND	Less than 80 mmHg	Healthy lifestyle habits, annual check
Elevated	120 - 129 mmHg	AND	Less than 80 mmHg	Lifestyle modification, re-check in 3-6 mo
Stage 1 HTN	130 - 139 mmHg	OR	80 - 89 mmHg	Lifestyle + ASCVD risk (meds if risk >=10%)
Stage 2 HTN	140 mmHg or higher	OR	90 mmHg or higher	Lifestyle + 2 first-line medications
Crisis	Higher than 180 mmHg	AND/OR	Higher than 120 mmHg	Immediate emergency medical evaluation

2. The Golden Rule of Category Assignment

* If systolic and diastolic values fall into different categories, the HIGHER category always determines sta
* Example: A reading of 124/86 mmHg is classified as STAGE 1 HYPERTENSION (due to diastolic >=80).
* Example: A reading of 144/78 mmHg is classified as STAGE 2 HYPERTENSION (due to systolic >=140).

3. Hypertensive Crisis: Urgency vs. Emergency Distinction

A. Hypertensive Urgency (Severe Elevation WITHOUT Target Organ Damage):

* SBP >180 mmHg and/or DBP >120 mmHg with NO acute symptoms of organ failure.
* Action: Rest for 5 minutes, repeat measurement. If sustained, contact physician within 24-48 hours.
* Does not typically require ICU admission; managed with oral antihypertensive regimen adjustments.

B. Hypertensive Emergency (Severe Elevation WITH Target Organ Damage):

* SBP >180 mmHg and/or DBP >120 mmHg WITH acute, life-threatening target organ damage.
* Symptoms: Crushing chest pain, severe dyspnea, facial droop, dysarthria, focal weakness, blurred vision.
* Action: IMMEDIATE 911 / 997 EMERGENCY CALL. Requires ICU admission and titratable intravenous therapy.

4. Standardized 6-Step Home Blood Pressure Monitoring (HBPM) Protocol

1. 30-Minute Pre-Test Abstinence: No caffeine (coffee/tea/energy drinks), tobacco, or vigorous exercise.
2. Bladder Emptying: Fully empty bladder prior to reading (full bladder adds 10-15 mmHg artifact).
3. 5-Minute Seated Rest: Sit quietly in a comfortable chair in a quiet room; do not speak or use smartphone.
4. Correct Anatomical Posture: Back supported, feet flat on the floor (uncrossed; crossing adds 5-8 mmHg).
5. Arm & Cuff Alignment: Arm supported on a table so that the cuff center is at EXACT HEART LEVEL.
6. Bare Skin & Proper Cuff Size: Place cuff on bare upper arm; small cuff overestimates BP by up to 10 mmHg.

7. Dual Measurement: Take 2 readings spaced 1 minute apart; log the average in a morning/evening diary.

5. White-Coat Hypertension and Masked Hypertension

* White-Coat Hypertension: Clinic BP >=140/90 mmHg but out-of-office BP normal (<130/80 mmHg).
* Masked Hypertension: Clinic BP normal (<130/80 mmHg) but out-of-office BP elevated (>=130/80 mmHg).
* Diagnostic confirmation requires 24-hour Ambulatory Blood Pressure Monitoring (ABPM).